Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
1	(a)		$l = \frac{RA}{\rho}$ [rearrangement at any stage] (1) $A = \frac{\pi d^2}{4}$ [$= 1.77 \times 10^{-8}$] or equivalent or by implication (1) $l = 2.0$ [m] (1)		3		3	3	
	(b)		Taking readings at face value, $R = \left[\frac{1.60}{0.029}\right] = 55$ [Ω] or 55.2 [Ω] **Or** lowest possible resistance =$\left[\frac{1.59}{0.030}\right] = 53.0$ [Ω] or 53 [Ω] (1) But highest possible resistance =$\left[\frac{1.61}{0.028}\right] = 58$ [Ω] or 57.5 [Ω] **or** calculating absolute uncertainty e.g. ± 2 [Ω] (1) accept 4% [56 Ω is between 55.2 Ω (or 53.0 Ω) and 57.5 Ω] so her conclusion is *not* safe (1) [Mark **not** freestanding] Award a max of 2 marks for upper values of V and I used along with lower values of V and I **Alternative:** Calculating V with 3 different possible I values and 56 [Ω] (2) Her conclusion is *not* safe (1) [Mark **not** freestanding]			3	3	2	3
			Question 1 total	**0**	**3**	**3**	**6**	**5**	**3**